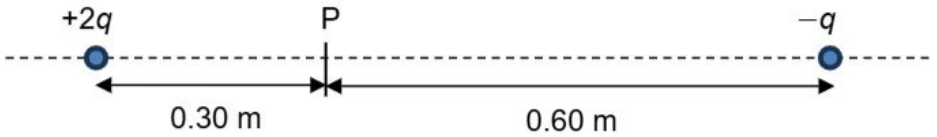
20 Two point charges $+2q$ and $-q$ are situated along a line at a distance 0.90 m apart. A point P is at a distance 0.30 m from the charge $+2q$.



The magnitude of the electric field strength at P due to the $+2q$ charge alone is 24 N C^{-1}.

What is the magnitude of the resultant electric field strength at P due to both charges?

	A	18 N C^{-1}	**B**	21 N C^{-1}	**C**	27 N C^{-1}	**D**	30 N C^{-1}

L2 **Answer: C**

Magnitude of Electric field strength due to +2q,

$$E_{2q} = \frac{2q}{4\pi\varepsilon_o\ 0.30^2} = 24 \text{ N C}^{-1}$$ **to the right**

Magnitude of Electric field strength due to −q,

$$E_q = \frac{q}{4\pi\varepsilon_o\ 0.60^2} = \frac{1}{8}\left(\frac{2q}{4\pi\varepsilon_o\ 0.30^2}\right) = \frac{1}{8}\ 24 = 3 \text{ N C}^{-1}$$ **to the right**

Resultant Electric field strength due to both +2q and −q = 24 + 3 = 27 N C^{-1} to the right